

AI Compliance Pre-Screening Pipeline

Case study — automated content generation and multi-layer AI/rule-based compliance review, built for a regulated-industry marketing workflow.

The Challenge

Marketing content in regulated industries (life sciences, finance, and similar) must pass a formal compliance review before it can be published — checking for unsupported claims, missing legal disclosures, invented statistics, non-compliant incentives, and unbalanced benefit statements. This review is typically slow, manual, and repeated for every piece of content produced.

Requirement: build an automated pipeline that generates on-brand marketing copy and immediately screens it against a defined set of compliance rules, producing a clear, evidence-based report a human reviewer can act on — reducing manual review time without removing human sign-off.

What I Built

An end-to-end, no-code/low-code automation workflow that:

- Generates coordinated marketing copy (e.g. email + landing page) from a creative brief and an approved source-of-truth document.
- Runs the generated copy through a dual-layer compliance check: deterministic rule matching (exact wording, formatting, required disclosures) and AI-based semantic review (unsupported claims, topic accuracy, invented statistics, non-compliant incentives, fair-balance requirements).
- Merges both layers into one structured report: rule-by-rule status, plain-language explanation, cited evidence, and a suggested correction.
- Supports live content updates (e.g. changed event details) with automatic re-verification.
- Outputs a shareable, downloadable compliance report for human reviewer sign-off.

Technical Highlights

- **Rule-engine design:** separated deterministic checks from AI-judged checks and merged results safely, eliminating a rule cross-contamination defect found and fixed during QA.
- **LLM reliability engineering:** diagnosed specific, repeatable blind spots in AI-based review (e.g. under-flagging subtly-framed fabricated statistics) and closed them with targeted prompt refinement — validated against false positives before shipping.
- **Self-consistency safety net:** implemented a dual independent AI review pass with a consensus-merge step, converting single-pass inconsistency (a known limitation of LLM-based review) into a materially higher effective catch rate.
- **Systematic QA discipline:** built and executed a full regression matrix — baseline behavior, isolated rule checks, combined-violation cross-contamination tests, edge-case updates, and stability checks — with every fix verified across multiple repeated runs before being signed off.

Outcome

A working automated compliance pre-screening tool covering eight distinct regulatory rule categories, combining deterministic and AI-based verification, with sub-40-second turnaround per submission and a fully documented, evidence-based testing process — hardened through systematic reliability testing rather than shipped on a first pass.

Skills & Tools

Workflow Automation (n8n) · Prompt Engineering · LLM Integration (OpenAI API) · AI Reliability & Hallucination Mitigation · Regulatory/Compliance Automation · Database Integration (Supabase) · JavaScript (Custom Logic Nodes) · Webhook & API Integration · QA / Regression Testing · Automated Report Generation